

SAMPLE VARIANCE $\hat{\sigma}^2$

$\Rightarrow \bar{X}_i \sim \text{pop}(\mu, \sigma^2)$; Sample: $\bar{X}, \hat{\sigma}^2$ $\{ \bar{X}_1, \dots, \bar{X}_n \}$

$$\rightarrow \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (\bar{X}_i - \bar{X})^2 = \frac{1}{n} \sum_{i=1}^n (\bar{X}_i^2 - \bar{X}^2)$$

$$\rightarrow E(\hat{\sigma}^2) = \frac{1}{n} \sum_{i=1}^n E[\bar{X}_i^2 - \bar{X}^2] =$$

$$= \frac{1}{n} \sum_{i=1}^n E \left[\underbrace{(\text{Var}(\bar{X}_i))}_{\sigma^2} + \underbrace{E(\bar{X}_i)^2}_{\mu^2} - \left(\underbrace{\text{Var}(\bar{X})}_{\frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1}\right)} + \underbrace{E(\bar{X})^2}_{\mu^2} \right) \right] =$$

$$= \frac{1}{n} \sum_{i=1}^n \left[\sigma^2 - \frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right) \right] =$$

$$= \frac{\sigma^2}{n} \left[n - \left(1 - \frac{n-1}{N-1} \right) \right] = \frac{\sigma^2}{n} \frac{N(n-1)}{N-1}$$

$$\Rightarrow E \left(\underbrace{\frac{n(N-1)}{N(N-1)} \hat{\sigma}^2}_{\text{unbiased estimate of } \sigma^2} \right) = \sigma^2 \quad \text{Population variance}$$

unbiased estimate
of σ^2

$\rightarrow$ Relationship between σ^2 and $\hat{\sigma}^2$

$$\text{Var}(\bar{X}) = \frac{\hat{\sigma}^2}{n-1} \left(1 - \frac{n}{N} \right) = \hat{\sigma}^2 \frac{N-n}{N(n-1)} = \frac{\sigma^2}{n} \left(1 - \frac{n-1}{N-1} \right)$$

CALCULATE

UNKNOWN